Database Schema Design

Hasnahena Web Application

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Entity-Relationship design for
the Core Banking Ledger

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Document Control

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Revision History

Ver.	Date	Author	Changes
1.0	—	Md Saif Ahmed	Initial schema draft
2.0	—	Md Saif Ahmed	Added audit and donation tables
3.0	March 22, 2026	Md Saif Ahmed	Final release with indexes

1 Introduction

1.1 Purpose

This document outlines the high-level Entity-Relationship design for the Hasnahena Core Banking Ledger. The schema is designed to support:

- ACID-compliant financial transactions
- Double-entry bookkeeping principles
- Complete transparency and auditability
- Shariah-compliant operations (zero interest)

1.2 Database Selection

PostgreSQL was selected as the primary database due to:

- Strong ACID compliance required for financial operations
- Native UUID support for distributed-friendly primary keys
- Excellent JSON support for flexible metadata storage
- Robust transaction isolation levels
- Open-source with enterprise-grade reliability

1.3 Conventions

- All primary keys use `UUID` type
- Timestamps use `TIMESTAMPTZ` (with timezone)
- Monetary values use `DECIMAL(15,2)` for precision
- Soft deletes via `deleted_at` column where applicable
- All tables include `created_at` and `updated_at`

2 Core Entities

2.1 User (Member/Worker/Admin)

The central entity representing all system users with role-based differentiation.

Column	Type	Description
user_id	UUID	Primary Key
role	ENUM	MEMBER, WORKER, ADMIN
full_name	VARCHAR(255)	Full name
phone	VARCHAR(20)	Unique, primary contact
email	VARCHAR(255)	Optional, unique
nid_number	VARCHAR(20)	National ID, unique
password_hash	VARCHAR(255)	Hashed password
status	ENUM	PENDING, ACTIVE, SUSPENDED
minimum_deposit_met	BOOLEAN	Loan eligibility flag
must_change_password	BOOLEAN	First login flag
created_at	TIMESTAMPTZ	Record creation time
updated_at	TIMESTAMPTZ	Last update time

2.2 Nominee

Stores nominee information for each member (required for financial accounts in Bangladesh).

Column	Type	Description
nominee_id	UUID	Primary Key
user_id	UUID	FK → User
nominee_name	VARCHAR(255)	Full name
nominee_nid	VARCHAR(20)	National ID
relation	VARCHAR(50)	Relationship to member
phone	VARCHAR(20)	Contact number
created_at	TIMESTAMPTZ	Record creation time
updated_at	TIMESTAMPTZ	Last update time

2.3 Transaction (Ledger)

Core ledger table for all financial transactions using double-entry principles.

Column	Type	Description
tx_id	UUID	Primary Key
initiator_id	UUID	FK → User (requester)
processor_id	UUID	FK → User (worker), nullable
amount	DECIMAL(15,2)	Transaction amount
tx_type	ENUM	See Transaction Types
status	ENUM	See Transaction Status
reference_id	UUID	Related entity (loan, donation)
notes	TEXT	Optional notes
processed_at	TIMESTAMPTZ	When processed by worker
confirmed_at	TIMESTAMPTZ	When confirmed by member
created_at	TIMESTAMPTZ	Record creation time

Transaction Types: DEPOSIT, WITHDRAWAL, LOAN_DISBURSEMENT, LOAN_REPAYMENT, DONATION_ZAKAT, DONATION_SADAQAH, DONATION_OPERATING, DONATION_LENDING

Transaction Status: PENDING, PROCESSED, CONFIRMED, REJECTED, DISPUTED, REVERSED

2.4 Loan

Tracks loan applications, approvals, and repayment status.

Column	Type	Description
loan_id	UUID	Primary Key
borrower_id	UUID	FK → User
principal_amount	DECIMAL(15,2)	Approved loan amount
amount_repaid	DECIMAL(15,2)	Total repaid so far
installment_count	INTEGER	Number of installments
installment_amount	DECIMAL(15,2)	Per installment amount
purpose	TEXT	Loan purpose description
status	ENUM	See Loan Status
approved_by	UUID	FK → User (admin)
disbursed_at	TIMESTAMPTZ	Disbursal timestamp
due_date	DATE	Final repayment due date
created_at	TIMESTAMPTZ	Application timestamp

Loan Status: PENDING_GUARANTOR, PENDING_APPROVAL, APPROVED, ACTIVE, SETTLED, DEFAULTED, REJECTED

2.5 Guarantor

Links guarantors to loans (many-to-many with approval workflow).

Column	Type	Description
id	UUID	Primary Key
loan_id	UUID	FK → Loan
guarantor_id	UUID	FK → User
status	ENUM	PENDING, APPROVED, REJECTED
responded_at	TIMESTAMPTZ	Response timestamp
created_at	TIMESTAMPTZ	Request timestamp

2.6 Donation Fund

Tracks donation pool balances by category.

Column	Type	Description
fund_id	UUID	Primary Key
fund_type	ENUM	OPERATING, LENDING, ZAKAT, SADAQAH
balance	DECIMAL(15,2)	Current balance
updated_at	TIMESTAMPTZ	Last update time

2.7 Audit Log

Immutable, append-only log for all system actions.

Column	Type	Description
log_id	UUID	Primary Key
actor_id	UUID	User performing action
action_type	VARCHAR(50)	Action identifier
entity_type	VARCHAR(50)	Affected entity type
entity_id	UUID	Affected entity ID
old_value	JSONB	Previous state
new_value	JSONB	New state
ip_address	INET	Client IP address
user_agent	TEXT	Client user agent
created_at	TIMESTAMPTZ	Action timestamp

3 Security & Best Practices

3.1 Double-Entry Accounting

Every credit in a member's account has an offsetting debit from a central operational or pool account. This ensures:

- Complete traceability of every transaction
- Balance verification through debit/credit matching
- Regulatory compliance for financial audits

3.2 Immutability

- **No Updates on Processed Transactions:** Once a transaction reaches CONFIRMED status, UPDATE statements are prohibited at the application level
- **Reversals via Compensation:** Corrections are performed by appending a compensating transaction with negative amount, preserving the original record
- **Audit Log Protection:** The audit_log table has no UPDATE or DELETE permissions granted to application roles

3.3 Indexing Strategy

Index	Purpose
user(phone)	Login lookup
user(email)	Login lookup
user(nid_number)	Uniqueness enforcement
transaction(initiator_id)	Member transaction history
transaction(status)	Pending queue queries
loan(borrower_id)	Member loan lookup
loan(status)	Active loan queries
audit_log(actor_id)	User action history
audit_log(created_at)	Time-range queries

4 Appendix: Entity Relationship Diagram

The full ER diagram is maintained separately in the project documentation. Key relationships:

- **User** → **Nominee**: One-to-One (each member has one nominee)
- **User** → **Transaction**: One-to-Many (initiator)
- **User** → **Loan**: One-to-Many (borrower)
- **Loan** → **Guarantor**: One-to-Many
- **User** → **Guarantor**: One-to-Many (guarantor role)
- **User** → **Audit_Log**: One-to-Many (actor)